Dense Graphs with Small Clique Number

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Abstract

We consider the structure of K_r-free graphs with large minimum degree, and show that such graphs with minimum degree $\delta > (2r-5)n/(2r-3)$ are homomorphic to the join $K_{r-3} \vee H$ where H is a triangle-free graph. In particular this allows us to generalize results from triangle-free graphs and show that K_r-free graphs with such minimum degree have chromatic number at most $r+1$. We also consider the minimum-degree thresholds for related properties.

1 Introduction

Triangle-free graphs, and in particular those with large minimum degree, have been the focus of many papers. There are results about bounds on the chromatic number [16, 5], the existence of homomorphisms [6, 11, 12, 14], the existence of similarity sets [8], and bounds on the binding number [15, 17]. On the other hand, K_r-free graphs with large minimum degree are not as studied. There is, of course, the famous old result due to Andrásfai, Erdős and Sós that a K_r-free graph with sufficiently large minimum degree is homomorphic to K_{r-1} and hence $(r-1)$-colorable. However, apart from papers such as [3, 18], the structure of K_r-free graphs with minimum degree just below this have not been explored much.

In this paper we establish results on the structure of K_r-free graphs with large minimum degree. In particular, we provide minimum-degree conditions for

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the existence of large independent sets, the existence of homomorphisms, and for bounded chromatic number. The main tool is a result (Theorem 8) that if the minimum degree is sufficiently large, then the graph can be partitioned into a K_{r-1}-free and K_k-free part for any chosen $k \in \{2, \ldots, r-1\}$. From this a general partitioning result (Theorem 9) follows. However, we start by presenting the results for K_4-free graphs, since these are simpler and sharper.

It should be noted that after acceptance we were made aware of the Habilitation thesis of Stephan Brandt [4]. Several of our results are contained in that thesis, and in fact, improved upon. Indeed, that thesis is devoted to the topic of the structure of dense graphs with small clique number, and contains many interesting results!

1.1 Previous results

The most well-known result for the chromatic number χ of dense graphs in terms of the minimum degree δ is due to Andrásfai, Erdős, and Sós [2], and states the following.

Theorem 1 ([2]) *Let G be a K_r-free graph such that $\delta(G) > (3r-7)n/(3r-4)$. Then $\chi(G) \leq r-1$.*

Alon and Sudakov [1] bounded the number of edges that need to be removed to obtain a similar result for all sufficiently large H-free graphs (where by H-free we mean a graph without H as a subgraph, not necessarily induced).

Theorem 2 ([1]) *Let H be a fixed graph on n_H vertices with chromatic number $r+1 \geq 3$, suppose $\varepsilon > 0$ and let G be an H-free graph of sufficiently large order n (depending only on n_H and ε) with minimum degree $\delta(G) \geq ((3r-7)/(3r-4)+\varepsilon)\, n$. Then one can delete at most $O(n^{2-1/(4r^{2/3}n_H)})$ edges to make G r-colorable.*

For triangle-free graphs in particular, Erdős and Simonovits [7] raised the question about the chromatic number of "dense" triangle-free graphs, with "dense" taken to mean $\delta > n/3$. It is easy to show, and well known, that if $\delta > 2n/5$

then a triangle-free graph is bipartite. Thomassen [16] proved that the chromatic number of triangle-free graphs with $\delta \geq cn$ with $c > 1/3$ is bounded as a function of c. Recently, Brandt and Thomassé significantly strengthened this:

Theorem 3 ([5]) *Let G be a triangle-free graph such that $\delta(G) > n/3$. Then $\chi(G) \leq 4$.*

There are also results about the existence of homomorphisms. Häggkvist [11] showed that a triangle-free graph of minimum degree at least $3n/8$ is either bipartite or homomorphic to the 5-cycle. This result was extended by Jin [12] and Chen et al. [6], inter alia. Łuczak [14] provided a general result about the structure of triangle-free graphs with minimum degree more than $n/3$, which was made more precise in [5]:

Theorem 4 ([14, 5]) *For every $\varepsilon > 0$, there exists a set of triangle-free graphs $\mathcal{G}_\varepsilon$ on at most $1/\varepsilon$ vertices, such that for every triangle-free graph G with $\delta(G) > (1/3 + \varepsilon)n$, G is homomorphic to some graph $H \in \mathcal{G}_\varepsilon$.*

In a different direction, Häggkvist's result was generalized by Győri et al. [9, 10].

Earlier, the paper [8] showed that a maximal triangle-free graph with minimum degree at least $(n+2)/3$ must have two *similar* vertices, that is, nonadjacent vertices with the same neighborhoods. Indeed:

Theorem 5 ([8]) *Let G be a maximal triangle-free graph such that $\delta(G) > n/3$. Then there exists a similarity set of size $3\delta - n$.*

2 Dense K_4-free Graphs

2.1 Structural Result and Independent Nonneighborhoods

Our goal in this section is to determine a minimum-degree condition that forces structure on the nonneighborhood of a vertex in a maximum independent set, thereby establishing the structure of the whole graph. We use the notation $H(u) = V(G) - N(u)$; note that $u \in H(u)$.

Theorem 6 *Let G be a maximal K_4-free graph with $\delta(G) > 3n/5$. Then there exists a vertex u such that $H(u)$ is an independent set.*

PROOF. Let I be a maximum independent set of G. We claim that the result holds for any vertex $u \in I$. Note that $I \subseteq H(u)$.

Suppose that $H(u)$ is not independent. That is, I is not all of $H(u)$. Then, by the maximality of I, we can choose $s_1 \in I$ and $s_2 \in H(u) - \{u\}$ that are adjacent. So, consider the set $I' = N(s_1) \cap N(s_2) \cap N(u)$. Since G is K_4-free, this is an independent set.

Since G is maximal K_4-free, there exists a pair of adjacent vertices x_1 and x_2 in $N(u) \cap N(s_2)$. It follows that $N(x_1) \cap N(x_2)$ is an independent set contained in $H(u)$ and containing s_2. Thus, we have $|N(s_2) \cap H(u)| \leq (n - d(u)) - (2\delta(G) - n)$. At the same time $|N(s_1) \cap H(u)| \leq (n - d(u)) - |I|$.

We can then use these upper bounds to determine a lower bound on $|I'|$:

$$\begin{aligned}
|I'| &\geq |N(s_1) \cap N(u)| + |N(s_2) \cap N(u)| - |N(u)| \\
&\geq 2\delta(G) - |N(s_1) \cap H(u)| - |N(s_2) \cap H(u)| - d(u) \\
&\geq 2\delta(G) - [2(n - d(u)) - |I| - (2\delta(G) - n)] - d(u) \\
&\geq 5\delta(G) - 3n + |I|.
\end{aligned}$$

Since $\delta(G) > 3n/5$, this implies $|I'| > |I|$, a contradiction. ■

Theorem 7 *A maximal K_4-free graph G with $\delta(G) > 3n/5$ is the join of an independent set and a triangle-free graph Γ, where $\delta(\Gamma) > |\Gamma|/3$.*

PROOF. By Theorem 6, there exists a vertex u such that $H(u)$ is an independent set. Of course, the subgraph $\langle N(u) \rangle$ is K_3-free. Furthermore, as in Lemma 1.3 of [2], for any vertex $v \in \langle N(u) \rangle$ we have

$$\frac{d_{\langle N(u) \rangle}(v)}{|\langle N(u) \rangle|} \geq \frac{\delta(G) - (n - d(u))}{d(u)} \geq 2 - \frac{n}{\delta(G)} > 2 - \frac{5}{3} = \frac{1}{3}.$$

■

Brandt [4] showed (a) that the minimum-degree condition in Theorem 7 can be changed to allow equality, and (b) that this is best possible.

2.2 Applications

From the above theorem, one immediately obtains a threshold for the existence of a large independent set:

Corollary 1 *Any K_4-free graph G with minimum degree $\delta(G) > 3n/5$ has an independent set of size $n/3$.*

Proof. Clearly we may assume that G is maximal K_4-free. By Theorem 7, G is the join of an edgeless graph and a triangle-free graph Γ. If Γ is bipartite, then G is 3-colorable, and so the independence number is as desired.

So assume that Γ is not bipartite. It follows that its minimum degree $\delta(\Gamma) \leq 2|\Gamma|/5$. If we let a denote the size of the edgeless graph of G, it follows that $3n/5 < \delta(G) \leq a + 2(n-a)/5$, which implies that $a > n/3$. The result follows. ■

One can also bound from below the minimum degree in the graph Γ, giving specific homomorphism results. For example, recall that Häggkvist [11] proved that any triangle-free graph G such that $\delta(G) > 3n/8$ is homomorphic to a 5-cycle or bipartite. Extending this, we get that:

Corollary 2 *Any K_4-free graph G with minimum degree $\delta(G) > 8n/13$ is homomorphic to a 5-wheel or a triangle.*

Proof. We may assume G is a maximal K_4-free graph. Since $\delta(G) > 8n/13 > 3n/5$, by Theorem 7, there is some vertex u such $H(u) = V - N(u)$ is an independent set. Again, $\langle N(u)\rangle$ is triangle-free, and for any vertex $v \in \langle N(u)\rangle$ we have

$$\frac{d_{\langle N(u)\rangle}(v)}{|\langle N(u)\rangle|} \geq \frac{\delta - (n - d(u))}{d(u)} \geq 2 - \frac{n}{\delta} > \frac{3}{8}.$$

By Häggkvist's result, this implies that $\langle N(u)\rangle$ is homomorphic to the 5-cycle or bipartite. Thus G is homomorphic to a 5-wheel or a triangle. ■

This result is sharp, as can be seen taking the Möbius ladder on 8 vertices $\overline{C_8^2}$ and joining it with an independent set of five vertices. In the same way, other results about homomorphisms of dense triangle-free graphs can easily be extended to K_4-free graphs. Indeed, by the results of [5, 14, 12] mentioned earlier, one can obtain thresholds for bounded chromatic number, and for homomorphism to one of a finite family of graphs. To avoid repetition, we defer the proofs and discussion of these results to later, since that is identical for K_4-free graphs and for K_r-free graphs with $r \geq 5$. See Section 3.4.

We compare the various minimum-degree thresholds for K_4-free graphs in Table 1.

Structural Property	*Minimum Degree Threshold*
Similarity sets	$\geq \frac{1}{2}n, \ \leq \frac{3}{5}n,$
Contained in join of an independent set and a triangle-free graph	$\frac{3}{5}n$
Independent set of size $\frac{n}{3}$	$\frac{3}{5}n$
Homomorphic to some graph in a fixed finite set of graphs	$\frac{3}{5}n$
5-colorable	$\frac{3}{5}n$
4-colorable	$\frac{29}{48}n$
3-colorable or homomorphic to wheel W_5	$\frac{8}{13}n$
3-colorable	$\frac{5}{8}n$

Table 1: Thresholds on the minimum degree for structural properties of dense K_4-free graphs.

3 Dense K_r-free Graphs In General

In this section we generalize several of the results on K_4-free graphs to K_r-free graphs for $r \geq 5$.

3.1 Some Simple Lemmas

In order to generalize Theorem 6, we will need a few simple lemmas. The first lemma deals with the degree requirements to extend any K_m to a K_{m+1}.

Lemma 1 *If $\delta(G) > (m-1)n/m$, any K_m in G can be extended to a K_{m+1}.*

PROOF. Suppose for contradiction that some K_m cannot be extended. Then the degree out from vertices in the K_m is $m\delta - m(m-1)$, and the degree in from any other vertex is at most $m-1$. Therefore, $(n-m)(m-1) \geq m\delta - m(m-1)$, which implies $n(m-1) \geq m\delta$, a contradiction. ■

We use a simple counting idea repeatedly. So we formalize it as a lemma. Let X be a set and let $\mathcal{A} = \{A_i\}_{i=1}^k$ be a family of subsets of X. We define the *slack* of this set system, $\mathit{Slack}(\mathcal{A})$, as

$$\mathit{Slack}(\mathcal{A}) = \sum_{x \in S} \left[(k-1) - r(x)\right],$$

where $S \subseteq X$ is the set of elements that are contained in at most $k-2$ members of $\mathcal{A}$, and $r(x)$ is the number of members of $\mathcal{A}$ containing an element x.

Lemma 2 *Let $\mathcal{A} = \{A_i\}_{i=1}^k$ be a family of subsets of some t-element universe X such that $|A_i| \geq s$ for each $A_i \in \mathcal{A}$. Then the size of the intersection of all the sets in $\mathcal{A}$ is at least the following:*

$$|\bigcap \mathcal{A}| \geq k \cdot s - (k-1) \cdot t + \mathit{Slack}(\mathcal{A}) \geq k \cdot s - (k-1) \cdot t.$$

PROOF. Let $|\bigcap \mathcal{A}| = a$, and consider $\sum_{i=1}^k |A_i|$. Clearly $\sum_{i=1}^k |A_i| \geq ks$. By the definition,

$$\begin{aligned} \textstyle\sum_{i=1}^k |A_i| &= ak + (t-a)(k-1) - \mathit{Slack}(\mathcal{A}) \\ &= a - \mathit{Slack}(\mathcal{A}) + (k-1)t. \end{aligned}$$

The lower bound on a follows. ■

Next, we consider extending independent sets, by noting that any two nonadjacent vertices can be extended to a large independent set, as noted by Kleitman [13] in a review of [2].

Lemma 3 *For $r \geq 3$, in a maximal K_r-free graph G, any two nonadjacent vertices are in an independent set of size $(r-2)\delta(G) - (r-3)n$.*

PROOF. Let v and v' be nonadjacent vertices in a maximal K_r-free graph G. By the maximality of G, the addition of the edge (v, v') forms a K_r. That is, there is some set of $r-2$ vertices, say S, such that $S \subseteq N(v) \cap N(v')$ and the graph $\langle S \rangle$ induced by S is K_{r-2}. Let $I = \bigcap_{w \in S} N(w)$. Then I is an independent set, with $v, v' \in I$. Additionally, any vertex not in I is adjacent to at most $r-3$ vertices of S. Thus, by Lemma 2, $|I| \geq (r-2)\delta(G) - (r-3)n$. ■

Note that the size of the independent set is best possible. This is trivial in the triangle-free case. In general, consider the complete $(r-1)$-partite graph $G = \overline{K_{a_1}} \vee \left(\bigvee_{r-2} \overline{K_{a_2}}\right)$, with $a_1 \leq a_2$: this graph has $\delta = n - a_2$, and from the theorem, we get every two nonadjacent vertices in an independent set of the size $(r-2)\delta - (r-3)n = (r-2)(n-a_2) - (r-3)n = a_1$, which is sharp.

3.2 K_k-free Nonneighborhoods

Our goal in this section is to determine minimum-degree conditions that force structure on the nonneighborhood of a vertex in a maximum independent set. As before we use the notation $H(u) = V(G) - N(u)$. The following result extends Theorem 6 to $r > 4$.

Theorem 8 *Let $r \geq 4$, $k \in \{2, \ldots, r-2\}$ and let $\delta_{r,k} = (k(r-3)+1)/(k(r-2)+1)$. Let G be a maximal K_r-free graph with $\delta(G) > \delta_{r,k} n$. Then there exists a vertex u such that subgraph $\langle H(u) \rangle$ induced by its nonneighborhood $V - N(u)$ is K_k-free.*

PROOF. Let I be a maximum independent set of G. We claim that the result holds for any vertex $u \in I$. Note that $I \subseteq H(u)$.

We will use induction on decreasing k. That is, since the expression $\delta_{r,k}$ increases as k grows smaller, we may assume the following: if $k < r-2$ that for all $u \in I$ that $H(u)$ is K_{k+1}-free, and we want to show that it is additionally K_k-free.

We assume for contradiction that there does exist some $K_k \subseteq H(u)$, say given by $\langle T \rangle$. From our minimum degree condition, we can apply Lemma 1 and extend this set T (if needed) to a set of vertices S such that $\langle S \rangle = K_{r-2}$.

Then, we consider the set $I' = \bigcap_{s_i \in S}(N(s_i) \cap N(u))$. Since G is K_r-free, this is an independent set. Since $|N(s_i) \cap N(u)| \geq \delta(G) - |N(s_i) \cap H(u)|$ and $|N(u)| = d(u)$ (the degree of u), by Lemma 2, the size of this set I' is bounded below:

$$|I'| \geq (r-2)\delta(G) - \left(\sum_{s_i \in S} |N(s_i) \cap H(u)| \right) - (r-3)d(u). \tag{1}$$

We then consider two cases depending on whether the clique T contains a vertex of I or not.

Case 1: *There is a set $T \subseteq H(u)$ such that $\langle T \rangle = K_k$ and $T \cap I \neq \emptyset$.*

Every vertex $v \in T$ is nonadjacent to u. So, by Lemma 3, vertices u and v are together in an independent set (contained in $H(u)$) of size at least $(r-2)\delta(G) - (r-3)n$. Thus, we have $|N(v) \cap H(u)| \leq (n - d(u)) - ((r-2)\delta(G) - (r-3)n)$. If we let v^* be the vertex in $T \cap I$, then $|N(v^*) \cap H(u)| \leq (n - d(u)) - |I|$. For every other vertex $s_j \in S - T$, it trivially holds that $|N(s_j) \cap H(u)| \leq |H(u)| = n - d(u)$.

We can then use these upper bounds and Inequality (1) to determine a lower bound on $|I'|$:

$$\begin{aligned}
|I'| &\geq (r-2)\delta(G) - \left[\sum_{s_i \in S} |N(s_i) \cap H(u)| \right] - (r-3)d(u) \\
&\geq (r-2)\delta(G) - [(r-2)(n - d(u)) - |I| - (k-1)\left((r-2)\delta(G) - (r-3)n\right)] \\
&\qquad - (r-3)d(u) \\
&\geq (r-2)\delta(G) + \delta(G) - (r-2)n + (k-1)\left((r-2)\delta(G) - (r-3)n\right) + |I| \\
&= [k(r-2)+1]\,\delta(G) - [k(r-3)+1]\,n + |I|.
\end{aligned}$$

Since $\delta(G) > \delta_{r,k} n$, this implies $|I'| > |I|$, a contradiction.

Case 2: *There is no set $T \subseteq H(u)$ such that $\langle T \rangle = K_k$ and $T \cap I \neq \emptyset$.*

By the induction hypothesis, if $k < r-2$ then we may assume that for all $u \in I$ that $H(u)$ is K_{k+1}-free. We now show that one can also assume this condition when $k = r-2$.

If $k = r-2$, assume for contradiction that $\langle H(u) \rangle$ is not K_{r-1}-free; i.e., there exists some set $U \subseteq H(u)$ such that $\langle U \rangle = K_{r-1}$. Removing some vertex v from the set U, we consider the set $I'' = \bigcap_{s_i \in U - \{v\}} N(s_i)$. This is independent since G is K_r-free. Since there is no k-clique in $H(u)$ containing a vertex of I, it follows that each vertex of I is nonadjacent to at least two vertices of $U - \{v\}$. If we let $X = V(G)$ and $\mathcal{A} = \{N(s_i)\}_{s_i \in U - \{v\}}$, it follows that $\mathit{Slack}(\mathcal{A}) \geq |I|$. So, application of Lemma 2 yields the following bound on $|I''|$:

$$|I''| \geq (r-2)\delta(G) - (r-3)n + |I|.$$

This shows that $|I''| > |I|$, since $(r-2)\delta(G) - (r-3)n > 0$ from the minimum degree condition; this is a contradiction of the maximality of I. That is, $\langle H(u) \rangle$ is K_{r-1}-free, as required.

Hence, for all k, under the conditions of this case, each vertex in I can be adjacent to at most $k-2$ of the vertices of any $K_k \subseteq \langle H(u) \rangle$. Further, every vertex of $H(u)$ not in $I \cup T$ can only be adjacent to at most $k-1$ of the vertices in the clique S, since $\langle H(u) \rangle$ is K_{k+1}-free by our induction hypothesis. Therefore,

$$\begin{aligned}\sum_{s_i \in T} |N(s_i) \cap H(u)| &\leq (k-2)|I| + (k-1)(n - d(u) - |I| - k) \\ &< (k-1)\,(n - d(u)) - |I|.\end{aligned}$$

For any other vertex $s_j \in S - T$, it holds that $|N(s_j) \cap H(u)| \leq |H(u)| = n - d(u)$. This shows that $\sum_{s_i \in S} |N(s_i) \cap H(u)| \leq (r-3)(n-d(u)) - |I|$. We can then use these upper bounds and Inequality 1 to determine a lower bound on $|I'|$:

$$\begin{aligned}|I'| &\geq (r-2)\delta(G) - [(r-3)(n-d(u)) - |I|] - (r-3)d(u) \\ &\geq (r-2)\delta(G) - (r-3)n + |I|.\end{aligned}$$

Since $(r-2)\delta(G) - (r-3)n > 0$, this implies $|I'| > |I|$, a contradiction.

Therefore, we have shown that for any vertex $u \in I$, the subgraph induced by its nonneighborhood $\langle H(u) \rangle$ is K_k-free. ■

3.3 Structural Result

Now, Theorem 8 can be used inductively to prove the main structural result about K_r-free graphs (which generalizes Theorem 7).

Theorem 9 *Let $r \geq 3$, $2 \leq k \leq r-1$, and let $\delta_{r,k} = (k(r-3)+1)/(k(r-2)+1)$. Let G be a maximal K_r-free graph with $\delta(G) > \delta_{r,k} n$. Then $V(G)$ can be partitioned into $\{S_0, S_1, \ldots, S_{r-(k+1)}\}$ where each induced graph $\langle S_i \rangle$, $i > 0$, is K_k-free and the induced graph $\langle S_0 \rangle$ is K_{k+1}-free with minimum degree larger than $\left(\frac{k^2-2k+1}{k^2-k+1}\right)|S_0|$.*

Proof. For a fixed value for k, we proceed by induction on increasing r. We note that the case where $r = k+1$ is trivial. So we assume $r \geq k+2$, and that the result holds for K_{r-1}-free graphs. Consider a K_r-free graph G, where $\delta(G) > \left(\frac{k(r-3)+1}{k(r-2)+1}\right) n$. By applying Theorem 8, we know that there exists some vertex u such that $\langle H(u) \rangle$ is K_k-free.

Recall that the subgraph $\langle N(u) \rangle$ is K_{r-1}-free. Furthermore, as in Lemma 1.3 of [2], for any vertex $v \in \langle N(u) \rangle$ we have

$$\frac{d_{\langle N(u) \rangle}(v)}{|\langle N(u) \rangle|} \geq \frac{\delta(G) - (n - d(u))}{d(u)} \geq 2 - \frac{n}{\delta(G)} > 2 - \frac{k(r-2)+1}{k(r-3)+1} = \frac{k(r-4)+1}{k(r-3)+1}.$$

Thus we can apply the inductive hypothesis to $\langle N(u) \rangle$ to obtain the rest of the desired partition. ■

In the special case of $k = 2$, we obtain the following:

Corollary 3 *Let G be a maximal K_r-free graph with $r \geq 4$ and $\delta(G) > (2r-5)n/(2r-3)$. Then, G is the join of a triangle-free graph and a complete $(r-3)$-partite graph; in particular, $G = \Gamma \vee_{j=1}^{r-3} \overline{K_{a_j}}$ for some a_j, where Γ is a triangle-free graph with minimum degree larger than $|\Gamma|/3$.*

Proof. Applying the previous theorem, we get a partition of G into independent sets $\overline{K_{a_j}}$ and a triangle-free graph Γ. And, since the graph $\Gamma \vee_{j=1}^{r-3} \overline{K_{a_j}}$ is a maximal K_r-free graph containing G, then $G = \Gamma \vee_{j=1}^{r-3} \overline{K_{a_j}}$. ■

Brandt [4] showed that one can allow equality in the minimum degree condition and get a similar result to Corollary 3.

3.4 Consequences of Theorem 9

For both of the below extensions, we simply apply the theorems of Brandt and Thomassé [5] and Łuczak [14] mentioned in Section 1.1 to the triangle-free subgraph Γ obtained in Corollary 3.

Theorem 10 *For every $\varepsilon > 0$, there exists a constant $M(\varepsilon)$ such that every K_r-free graph with $\delta(G) \geq ((2r-5)/(2r-3)+\varepsilon)\,n$, is homomorphic to a K_r-free graph on at most $M(\varepsilon)$ vertices.*

Theorem 11 *Let G be a K_r-free graph with $\delta(G) > (2r-5)n/(2r-3)$. Then $\chi(G) \leq r+1$.*

The degree bound for Theorems 10 and 11 are sharp. For, we can define a class of graphs that are K_r-free and have minimum degree approaching $(2r-5)n/(2r-3)$ such that the sequence has unbounded chromatic number. This is achieved by exploiting the construction of Erdős, Hajnal, and Simonovits [7]. For any k and $\varepsilon > 0$, they constructed a triangle-free graph $G(k,\varepsilon)$ such that $\chi(G(k,\varepsilon)) \geq k$ and $\delta(G(k,\varepsilon)) \geq (1/3-\varepsilon)n_{k,\varepsilon}$, where $n_{k,\varepsilon}$ is the number of vertices in $G(k,\varepsilon)$. We define the graph $H(r,k,\varepsilon) = G(k,\varepsilon) \vee_{i=1}^{r-3} \overline{K_\ell}$, where $\ell = 2n_{k,\varepsilon}/3$. This graph has $n = (2r/3-1)n_{k,\varepsilon}$ vertices, chromatic number at least $k+r-3$, and minimum degree larger than $((2r-5)/(2r-3)-\varepsilon)n$.

Theorem 11 is also sharp in that there are K_r-free graphs with minimum degree more than $(2r-5)n/(2r-3)$ that have chromatic number $r+1$, just as there are in the $r=3$ case. Indeed, Jin [12] showed that if the minimum degree is more than $10n/29$, then a triangle-free graph is 3-colorable. This is best possible due to the graph F_{29m} constructed by Häggkvist [11] (the Grötzsch graph with each vertex replaced by some number of vertices). Using this, one can readily obtain the corresponding results for general K_r-free graphs:

Theorem 12 *Let G be a K_r-free graph with $\delta(G) > (19r-47)n/(19r-28)$. Then $\chi(G) \leq r$.*

Proof. The proof is by induction on r, with the base case $r=3$ given by Jin [12]. We may assume G is a maximal K_r-free graph. Since $\delta(G) > (2r-5)n/(2r-3)$, by Corollary 3, there is some vertex u such that the induced subgraph $H(u) = V - N(u)$ is independent. At the same time, $\langle N(u)\rangle$ is K_{r-1}-free, and for any vertex $v \in \langle N(u)\rangle$ we have

$$\frac{d_{\langle N(u)\rangle}(v)}{|\langle N(u)\rangle|} \geq \frac{\delta-(n-d(u))}{d(u)} \geq 2-\frac{n}{\delta} > \frac{19(r-1)-47}{19(r-1)-28}.$$

Thus, $\chi(\langle N(u)\rangle) \leq r-1$, and the result follows. ■

The minimum-degree threshold is best possible, as can be shown by taking the extremal graph F_{29m} for the triangle-free case, and joining it with a suitable complete $(r-3)$-partite graph. Similar result are obtained by Brandt [4].

One can also bound from below the minimum degree in the graph Γ, giving specific homomorphism results, as we did for K_4-free graphs.

3.5 More on the Independence Number

It is clear that once the K_r-free graph has chromatic number $r-1$, then it has an independent set of cardinality $n/(r-1)$; and this is the largest that one can ever force via the minimum degree. In triangle-free graphs, the minimum-degree thresholds where χ becomes 2 and where there is guaranteed an independent set of size $n/2$ are the same, namely $2n/5$. However, we showed that for $r=4$, the threshold guaranteeing independence number $n/3$ (the maximum one can force) is lower than the threshold that guarantees chromatic number at most 3 (the minimum one can force).

For $r \geq 5$, one can similarly show using Corollary 3 that minimum degree more than $(2r-5)n/(2r-3)$ implies a K_r-free graph has an independent set of size at least $n/(r-1)$. But the threshold in this case actually lies below this minimum degree. To show this, we will need the following result, which improves on Corollary 3:

Theorem 13 *Let G be a maximal K_r-free graph, $r \geq 5$, with $\delta(G) > (r-3)n/(r-2)$. Then either G has an independent set I, such that*

$$|I| \geq 2\left[(r-2)\delta(G)-(r-3)n\right],$$

or G is the join of an independent set and a K_{r-1}-free graph.

Proof. Let I be a maximum independent set in G. If every other vertex is adjacent to all of I, then G is the join of an independent set and a K_{r-1}-free graph, so we are done. Thus we may assume that there exists a vertex

$w \in V(G) - I$ such that w is not adjacent to all of I (but by the maximality of I is adjacent to some of I).

Consider some vertex $x \in I$ so that x is nonadjacent to w. Then, by the maximality of G, there must exist some K_{r-2} in $N(x) \cap N(w)$. Let the set S be those $r-2$ vertices along with w. Note that $\langle S \rangle = K_{r-1}$, but $S \cap I = \emptyset$. For each $s_i \in S$, we define $k_i = |I - N(s_i)|$.

Now, we will consider three cases, based on the values of the k_i. In each case, we will build a set T, so that the graph induced by T is K_{r-2}. In anticipation of applying Lemma 2, we define $\mathcal{A} = \{N(t_i)\}_{t_i \in T}$ and consider bounds for $\mathit{Slack}(\mathcal{A})$.

1. *There exists some i such that $k_i \geq |I|/2$.*
   In this case, we take any edge from s_i to a vertex y in I, and using Lemma 1, extend the pair to a set of vertices T such that the graph induced by T is K_{r-2}. Then at least k_i vertices in I are nonadjacent to 2 vertices in T (namely s_i and y), which implies $\mathit{Slack}(\mathcal{A}) \geq |I|/2$.

For the remaining two cases, we may assume that any two vertices in S share a common neighbor in I.

2. *There exists some i, j $(i \neq j)$ such that $k_i + k_j \geq |I|/2$.*
   In this case, let y be any vertex in I which is adjacent to both s_i and s_j. Using Lemma 1, we can extend (if needed) $\{s_i, s_j, y\}$ to a set of vertices T such that the graph induced by T is K_{r-2}. Since all vertices in I are nonadjacent to y, and s_i and s_j have at least $k_i + k_j$ nonadjacencies in I, this implies that $\mathit{Slack}(\mathcal{A}) \geq k_i + k_j \geq |I|/2$.

3. *There is no pair i, j such that $k_i + k_j \geq |I|/2$.*
   Let $s_i = w$ and let s_j be any other vertex of S. Then let $T = (S - \{s_i, s_j\}) \cup \{x\}$. Every vertex $z \in I$ that is adjacent to both w and s_j must be nonadjacent to some vertex $s_i \in S - \{w, s_j\}$, otherwise we have a K_r. And there are at least $|I| - k_i - k_j$ choices for z. It follows that $\mathit{Slack}(\mathcal{A}) \geq |I|/2$.

We can then consider the independent set I', where $I' = \bigcap_{t_i \in T} N(t_i)$. Applying Lemma 2 with the bounds obtained on the quantity $\mathit{Slack}(\mathcal{A})$, we get

$|I| \geq |I'| \geq (r-2)\delta(G) - (r-3)n + |I|/2$. Solving for $|I|$, we get $|I| \geq 2\left[(r-2)\delta(G) - (r-3)n\right]$. ■

Theorem 14 *Let G be a maximal K_r-free graph, $r \geq 5$, with $\delta(G) > \frac{2r^2-8r+7}{2r^2-6r+4}n$. Then G has an independent set of size at least $\frac{n}{r-1}$.*

PROOF. First, suppose that G cannot be partitioned into an independent set and the neighborhood of a vertex. Then by Theorem 13, G must contain an independent set of size at least $2\left[(r-2)\delta(G) - (r-3)n\right]$. Plugging in a minimum degree of $\delta(G) > \frac{2r^2-8r+7}{2r^2-6r+4}n$ yields an independent set of size at least $n/(r-1)$.

In the case that G can be partitioned into an independent set and the neighborhood of a vertex, we proceed by induction on r. The degree of a vertex, say u, in the independent set is at least $\frac{r-2}{r-1}n$, or the result is trivial. Considering the minimum degree of the K_{r-1}-free graph induced by this neighborhood, we get that

$$\frac{d_{\langle N(u)\rangle}(v)}{|\langle N(u)\rangle|} \geq \frac{\delta(G) - (n - d(u))}{d(u)} \geq \frac{r-1}{r-2}\frac{\delta(G)}{n} - \frac{1}{r-2}.$$

For the case $r = 5$, then

$$\frac{r-1}{r-2}\frac{\delta(G)}{n} - \frac{1}{r-2} = \frac{4}{3}\left(\frac{17}{24}\right) - \frac{1}{3} = \frac{11}{18} > \frac{3}{5}.$$

From Corollary 1, there is an independent set of size at least $|N(u)|/3$ contained in the K_4-free induced graph formed from the vertices in $N(u)$ (where $|N(u)| \geq \frac{3}{4}n$), and this proves the base case.

For the case $r > 5$, the following general expression is obtained:

$$\frac{r-1}{r-2}\frac{\delta(G)}{n} - \frac{1}{r-2} \geq \frac{2r^2 - 10r + 11}{2(r-2)^2} > \frac{2(r-1)^2 - 8(r-1) + 7}{2(r-1)^2 - 6(r-1) + 4}.$$

Therefore, the induction hypothesis can be applied to prove the result. ■

It should be noted that $\frac{2r^2-8r+7}{2r^2-6r+4}n < \frac{2r-5}{2r-3}n$ (the bound of Theorem 11). However, this bound is not best possible (except possibly for $r = 6$), and is improved upon by Brandt [4].

4 Conclusion

In conclusion, we have been able to extend many of the results from dense triangle-free graphs to dense K_r-free graphs. Several of the results and bounds seem a natural extension of their triangle-free counterparts, but there are still many questions which arise, especially as regards sharpness.

Finally, it is very annoying that we are unable to prove an equivalent of Theorem 5 even for K_4-free graphs. For example, we were unable to resolve whether it is true that a maximal K_4-free graph of minimum degree at least $n/2 + 1$ must have two similar vertices.

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